

Windows: UEFI Secure Boot & TPM

1. Enable UEFI Secure Boot

- Reboot into UEFI/BIOS settings (often via F2 or Delete at startup), navigate to **Security** → **Secure Boot**, and enable it; select Microsoft's default key database (DB).

2. Activate TPM-based key storage

- In Windows, run `tpm.msc`, check TPM status, and initialize if needed; then enable **BitLocker** under **Control Panel** → **System and Security** → **BitLocker Drive Encryption** to tie disk encryption keys to the TPM.

3. Configure Group Policy for Secure Boot

- Under **Computer Configuration** → **Administrative Templates** → **Windows Components** → **Secure Boot**, enforce platform integrity checks per CIS Benchmark Level 1 recommendations.

Example: On a managed Dell fleet, IT pre-configures desktops with TPM-backed BitLocker and enforces Secure Boot via SCCM, preventing unauthorized OS loaders.

macOS: Apple's Boot Protection

1. Enable FileVault encryption

- Open **System Settings** → **Privacy & Security** → **FileVault** and click "Turn On" to encrypt the startup disk, leveraging the T2/Apple Silicon security chip as root of trust.

2. Verify SIP (System Integrity Protection)

- SIP blocks unauthorized modifications to system files. In **Terminal**, run `csrutil status`; it should report "enabled".
Note: Disabling SIP or FileVault undermines macOS's boot-chain protections and should only be done for very specific troubleshooting under expert guidance.

Disabling unnecessary services and features

Every running service or enabled feature on your device expands its attack surface. By turning off what you don't need, you reduce potential entry points for attackers.

Android: Strip down to essentials

1. Review and disable bloatware

- Navigate to **Settings** → **Apps & notifications** → **See all apps**.
- Select preinstalled or rarely used apps, tap **Disable** to prevent them from running in the background.

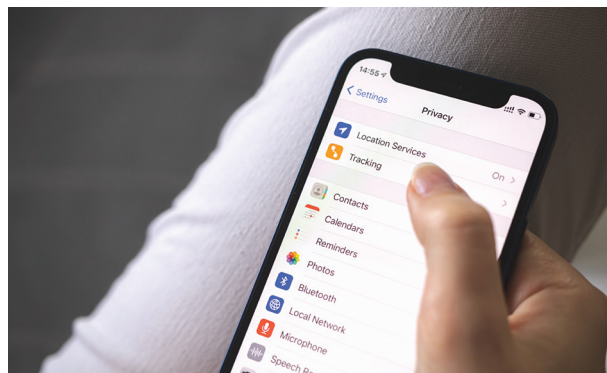
2. Turn off developer options unless in use

- **Settings** → **System** → **Developer Options** → **Toggle Off**. This prevents accidental USB debugging or OEM unlocking.

3. Disable unused radios

- Swipe down the quick-settings shade and toggle Bluetooth, NFC, or Location off when not in active use.
Tip: Use ADB to fully uninstall persistent bloatware on supported devices:

```
adb shell pm uninstall --user 0 com.vendor.unwantedapp
```



iOS: Lean and locked down

1. Disable background app refresh

- **Settings** → **General** → **Background App Refresh** → **Off** or selectively disable per app.

2. Limit cellular data

- **Settings** → **Cellular** → Scroll to apps → **Toggle Off** for any that don't need network access.

3. Disable Siri and Widgets on lock screen

- **Settings** → **Face ID & Passcode** → Scroll to "Allow Access When Locked" → **Toggle off** Siri, Today View, and Control Center if you prefer tighter lockdown.

Windows & macOS: Service Hardening

1. Windows: Turn off unneeded features

- **Control Panel** → **Programs** → **Turn Windows features on or off** → Disable features like SMBv1, Telnet, Legacy Components unless explicitly needed.
- **Services.msc**: Set non-essential services (e.g., Remote Registry) to **Manual** or **Disabled**.

2. macOS: System Preferences cleanup

- **System Settings** → **General** → **Login Items** → Remove apps you don't need launching at startup.
- **Sharing Preferences**: Turn off **File Sharing**, **Remote Login**, and **Screen Sharing** if unused.

Certificate pinning and App-level controls

Certificate pinning binds your application to a specific server